We hope you can appreciate that for some of those examples you could make arguments for or against it having the Markov property. For example, in diagnosing a patient, you may need to know the recent history of their symptoms, but if that is documented in their medical record and we consider the full medical record as our current state, then we've effectively induced the Markov property. This is an important thing to keep in mind: many problems may not *naturally* have the Markov property, but often we can induce it by jamming more information into the state.

DeepMind's deep Q-learning (or deep Q-network) algorithm learned to play Atari games from just raw pixel data and the current score. Do Atari games have the Markov property? Not exactly. In the game Pacman, if our state is the raw pixel data from our current frame, we have no idea if the enemy a few tiles away is approaching us or moving away from us, and that would strongly influence our choice of actions to take. This is why DeepMind's implementation actually feeds in the last four frames of gameplay, effectively changing a non-MDP into an MDP. With the last four frames, the agent has access to the direction and speed of all players.

Figure 2.8 gives a lighthearted example of a Markov decision process using all the concepts we've discussed so far. You can see there is a three-element state space `S = {crying baby, sleeping baby, smiling baby}`, and a two-element action space `A = {feed, don't feed}`. In addition, we have transition probabilities noted, which are maps from an action to the probability of an outcome state (we'll go over this again in the next section). Of course, in real life, you as the *agent* have no idea what the transition probabilities are. If you did, you would have a *model* of the environment. As you'll learn later, sometimes an agent does have access to a model of the environment, and sometimes not. In the cases where the agent does not have access to the model, we may want our agent to learn a model of the environment (which may just approximate the true, underlying model).

Figure 2.8. A simplified MDP diagram with three states and two actions. Here we model the parenting decision process for taking care of an infant. If the baby is crying, we can either administer food or not, and with some probability the baby will transition into a new state, and we'll receive a reward of -1, +1, or +2 (based on the baby's satisfaction).